AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I031*

Team Name: *Fantastic Four*

Domain:*Space,Aerospace,Mobility and Next-Gen Infrastructure*

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Problem Statement

How can we prevent defunct satellites from becoming long-term space debris and how can we safely manage or remove satellites that have already reached the end of their operation lives?

Problem Understanding

- The space agencies and organizations that launch satellites—like NASA, ESA, ISRO, Roscosmos State Corporation for Space Activities and private companies like SpaceX. Satellite operators like telecom, broadcasting, GPS companies, they need safe orbits for their assets.

- Causes includes fuel depletion, propulsion failure, communication loss, and degradation of onboard components. Many satellites lack reliable backup mechanisms to ensure safe disposal when critical systems fail. High removal costs and limited in-orbit servicing make safe satellite retirement more challenging.

- Space debris affects secondary stakeholders such as the public, governments, airlines, scientists, and space businesses by disrupting satellite-based services, increasing operational costs, threatening space missions, and creating potential safety risks. It also increases insurance costs and limits future access to valuable orbits. DDD aims to reduce these risks by removing selected debris and promoting a safer, more sustainable space environment.

Existing Gaps & Challenges

- Existing methods are piecemeal, expensive, and not universally enforced.Small debris and legacy junk remain unaddressed. No scalable, cooperative global framework exists.

- Limitations include the requirement of additional fuel or hardware, increasing the satellite's mass and overall mission cost. These methods are not suitable for every orbit or satellite, while active debris removal is complex and expensive. Failed or unresponsive satellites are difficult to manage and safely remove from orbit.

- The cost effectiveness and the moral obligation to dispose the debris; In today's world, with the current technology available it is not an economically feasible option for space agencies to invest in orbital debris management. Since the early 2000's, there has been an exponential increase in the number of satellites and space probes being launched which inturn poses a threat of an amplified Kessler's Syndrome in the near future.

Our Proposed Solution

- DDD (Debris Disposal Drone) is a proposed semi-autonomous spacecraft designed to tackle space debris by detecting, capturing, and safely disposing of retired satellites and other trackable orbital debris. Equipped with robotic arms, autonomous navigation, orbital propulsion, and deployable drag sails, DDD would operate across designated orbital regions, with additional drones deployed as debris-removal demand increases.

- DDD operates through a network of autonomous debris-removal spacecraft strategically positioned across selected orbital regions. Initially, 8 drones can be deployed at 45° intervals along a designated orbit, with the fleet expanding to 12 drones at 30° intervals as debris-removal demand increases. Each drone identifies, captures, and safely disposes of suitable orbital debris, while additional drones improve service density and operational capacity. This scalable approach aims to reduce collision risks, support responsible satellite retirement, and promote long-term space sustainability.

- "Redefining satellite retirement, so yesterday's missions don't become tomorrow's obstacles."

Technical Approach



Debris Disposable Drone

Tech Stack

Frontend: React-based Mission Control Dashboard

Backend: Python, FastAPI

Database: PostgreSQL

AI/ML, APIs, or Hardware: Orbital debris tracking data, orbital trajectory simulation, AI-assisted target prioritisation, autonomous navigation, robotic capture arms, propulsion system, deployable drag sail and thermal protection system.

Innovation & Uniqueness

- Its scalable, distributed orbital-cleanup network, designed to move beyond isolated debris-removal missions. Starting with 8 drones at 45° spacing, the fleet can expand to 12 drones at 30° spacing as demand grows, improving service density across selected orbital regions. By combining reusable debris-removal spacecraft, strategic orbital positioning, and planned end-of-life satellite disposal, DDD aims to tackle existing debris while preventing future accumulation.
- DDD is suited to this problem because it tackles space debris at its source by actively removing hazardous objects instead of relying solely on collision avoidance. Its strategically positioned drones can prioritise debris that poses the greatest risk, while adjustable fleet density allows resources to be deployed where they are needed most. This makes orbital cleanup more targeted, adaptable, and practical for long-term space operations.
- The existing theoretical solution rely on changing the orbit of the space debris such that it does not obstruct future expeditions, whereas we, The Fantastic Four plan on bringing the satellite back to the Earth for recycling and research purposes.

Feasibility & Scalability

- **Feasibility:** Building DDD requires orbital simulation software such as Python and GMAT, CAD tools such as Fusion 360, debris-tracking data, robotic-arm components, sensors, microcontrollers, and a team with aerospace, mechanical and software skills.

- **Scalability:** The system can be scaled by adding more features, supporting larger amounts of data, and integrating new tools and technologies. It can also be expanded to support more users, handle increased workloads, and provide more accurate results as requirements grow.

- **Risks:** The main constraints for DDD include limited fuel for satellite deorbiting, high costs of debris removal, communication failures with inactive satellites, and challenges in tracking debris accurately. These can be mitigated by using drag-assisted deorbiting modules, cost-effective modular designs, predictive tracking systems, and automated retirement planning to improve satellite disposal and reduce orbital debris.

Cost for Product (Optional)

- Development or hardware cost: Rs.5,000–Rs.10,000 approximately
- Hosting or maintenance cost: Rs.500–Rs.1,000 per month
- Estimated total: Rs.6,000–Rs.11,000 for the first month

Use Cases (compulsorily 3 use cases must be there)

- Use case 1:Space Agencies — ISRO and other space agencies use DDD to monitor orbital debris and plan safer satellite operations.
- Use case 2: Satellite Operators — Satellite companies use DDD to identify collision risks and plan satellite manoeuvres to avoid debris.
- Use case 3: Debris Management Teams — Space sustainability teams use DDD to identify potential debris-removal targets and plan safe deorbiting operations.
- Use case 4: Telecommunication Services — Telecommunication companies use DDD to monitor debris around communication satellites, reduce collision risks, and help maintain uninterrupted services such as mobile networks, television broadcasting, and satellite internet.
- Use case 5: Space Surveillance — Surveillance agencies use DDD to track orbital debris, monitor satellite movements, and detect potential threats in space, helping improve space situational awareness and protect critical space assets.

Impacts and Benefits

- The target users of DDD include space agencies, satellite operators, telecommunication companies, surveillance agencies, and space debris management teams. They benefit from improved debris tracking, early collision warnings, safer satellite operations, fewer communication disruptions, better space surveillance, and more effective debris removal, helping protect valuable space assets and keep Earth's orbit safer.
- Prevents disruptions to satellite-based internet, GPS, television broadcasting, and emergency communication services caused by space debris collisions.Helps satellite operators reduce replacement costs, avoid service downtime, and protect revenue by identifying collision risks early.Reduces the accumulation of orbital debris through improved tracking, collision avoidance, and planned disposal of inactive satellites, supporting long-term space sustainability.
- DDD is expected to improve the accuracy of space-debris tracking, identify potential satellite collisions earlier, and reduce disruptions to communication and surveillance services. Its success can be measured through tracking accuracy, collision alerts generated, detection time, and the number of debris objects monitored and identified for removal.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 9	Target 9.4	Industry, Innovation and Infrastructure	The Debris Disposal Drone (DDD) uses innovative technology to capture and remove defunct satellites, helping create safer and more sustainable space infrastructure.
SDG 12	Target 12.5	Responsible Consumption and Production	DDD reduces orbital waste by collecting and removing end-of-life satellites.

Individual Member Contribution

Member	Contribution or responsibility
Jahnavi A Nair	Active team leader and gathering data from research papers, creative contribution in designing chart paper.
Aditi.R	Compiling information into the presentation and data verification.
Dushyant.R	Electronic resources and ideating proposed solution.
S.Aditya	Implementing overall idea into a feasible soltuion and compiling data.

Yukti ID

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